GNC Enabling Technologies for Space Vehicles

CEC 300: Computing in Aviation and Aerospace

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Outline

- Introduction
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  - Thrusters

- GNC-enabling Sensors
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  - Magnetometers
  - Sun Sensors
  - Horizon Sensors
  - Inertial Sensing
    - GPS Receivers
    - Deep Space Navigation
    - Atomic Clocks
    - LiDAR

Pointing Control - NASA Science

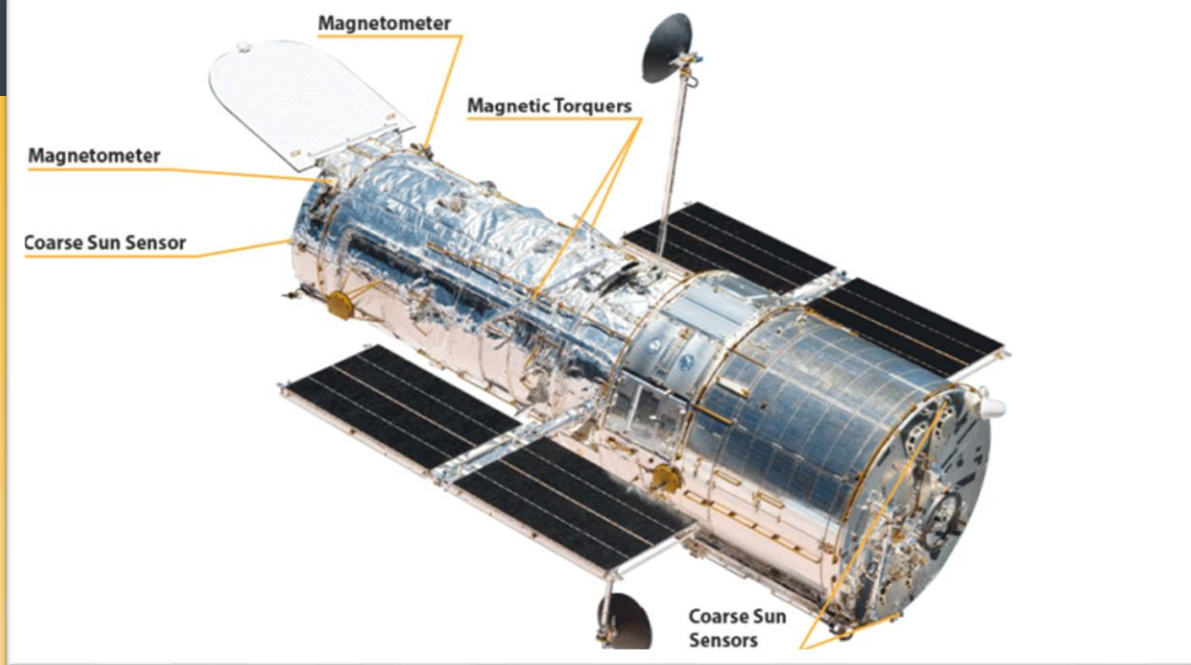
Introduction

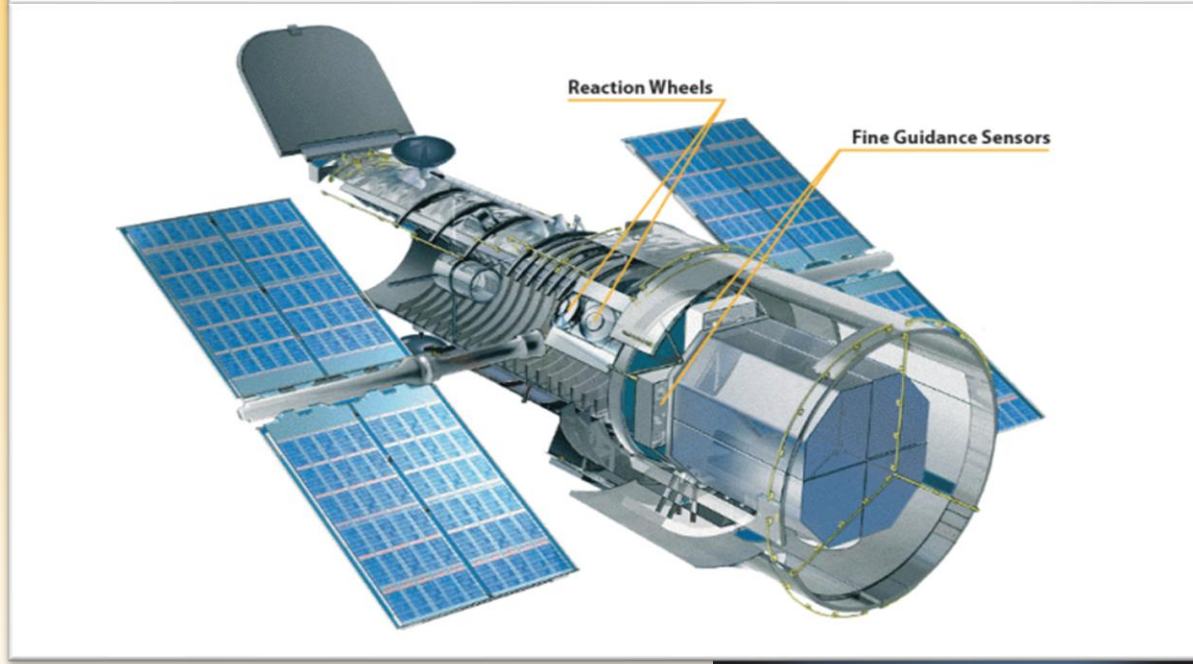
- Components of GNC for
  - Position
  - Attitude / Pointing control
  - Control System

- Position
  - Determines position of the spacecraft
    - Orbital Earth-centered inertial (ECI) frame
    - Deep space – International Celestial Reference Frame (ICRF)
      - Areas beyond cislunar space

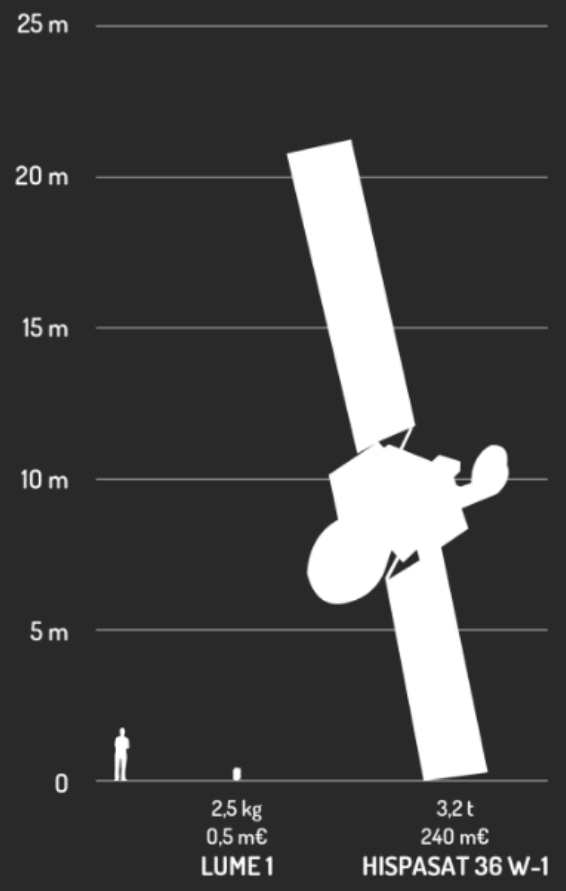
- Attitude and spin rate
  - GNC requires an understanding of the space vehicle's attitude and the rate at which it is turning about its axes.

- Actuators
  - Change attitude of aircraft and adjust velocities of space vehicle





Types of Satellites and their Scale



HOW BIG IS A NANOSATELLITE?

Artificial satellites vary in size and cost depending on the use they are put to. They can be small enough to fit in the palm of your hand or as huge as the ISS. According to NASA, “in terms of mass, a nanosat or nanosatellite is anything that weighs between 1 and 10 kilograms”.

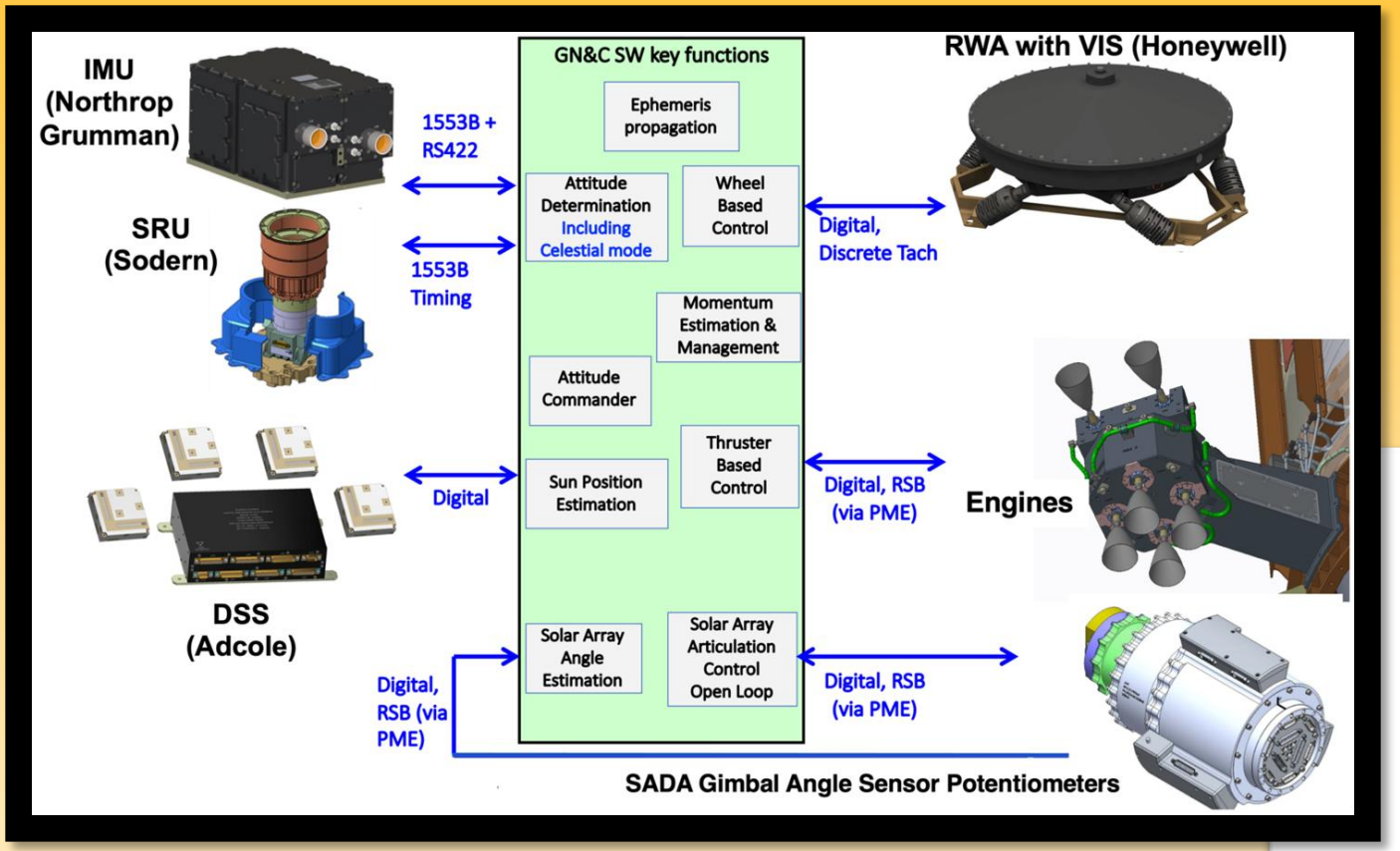
Satellite types according to mass:

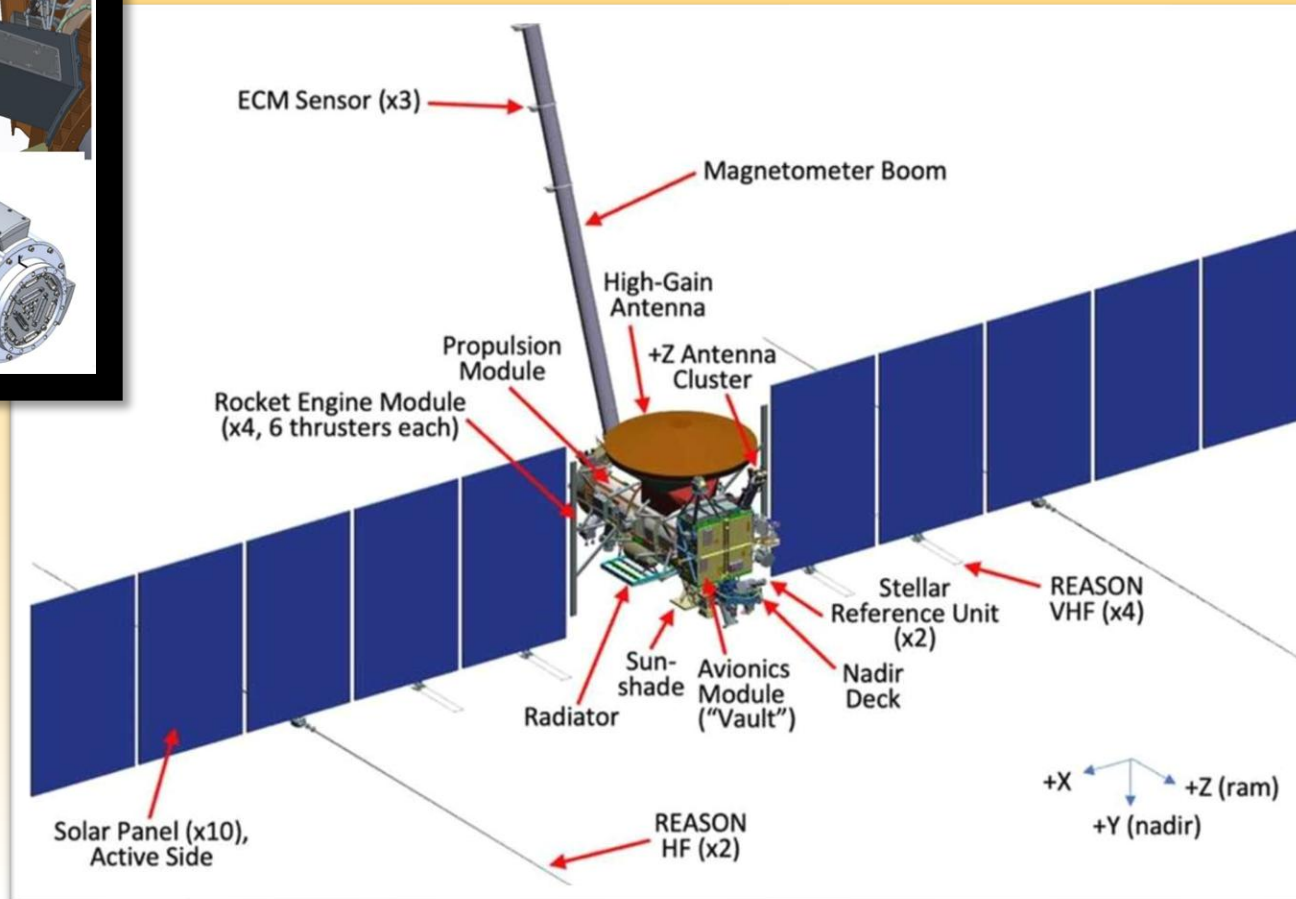
- Large satellites: More than 1,000 kg
- Medium-sized satellites: 500-1,000 kg
- Small satellites:
  - Minisatellite: 100-500 kg
  - Microsatellite: 10-100 kg
  - Nanosatellite: 1-10 kg
  - Picosatellite: Less than 1 kg

In recent years, standards have also been developed for smaller satellites such as femtosatellites, attosatellites, zeptosatellites and others.

A Basic Guide to Nanosatellites | Alén Space

Example: Europa Clipper



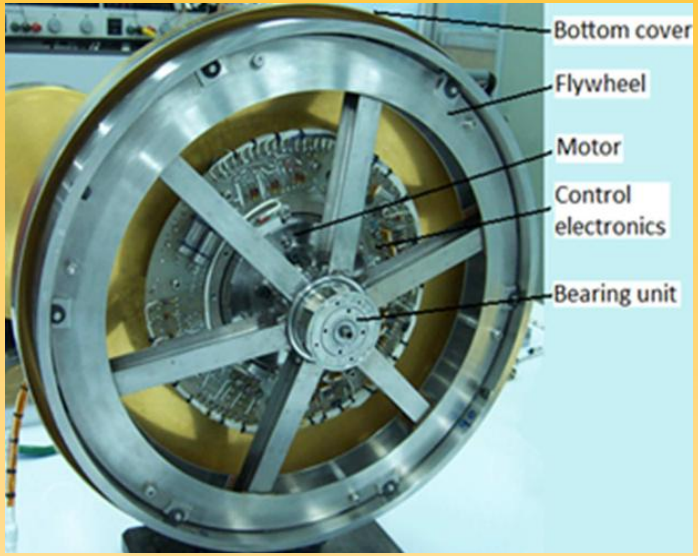


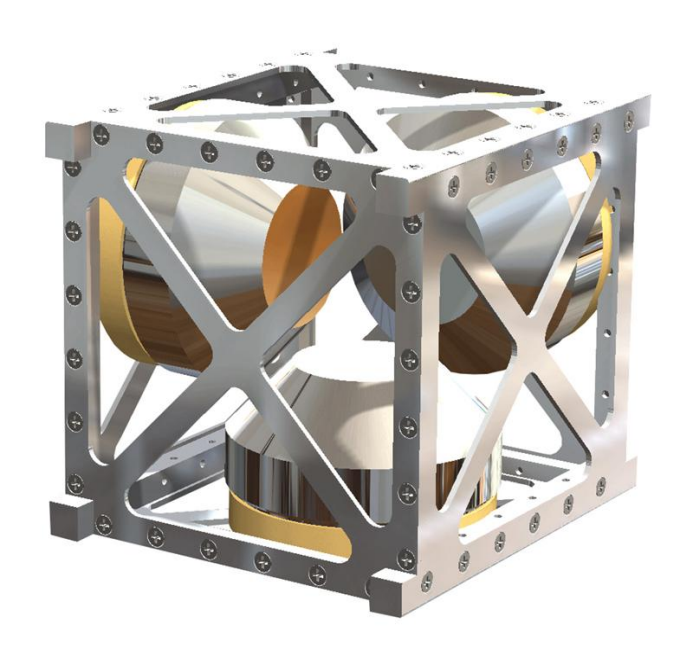
[Figure 3 | Europa Clipper Mission Design, Mission Plan, and Navigation | Space Science Reviews | Springer Nature Link]

Integrated Units

- Integrate multiple attitude and navigational components
- Typical components include:
  - Reaction Wheel
  - Magnetometer
  - Magnetic torquers
  - Fine and coarse sun sensors
  - GPS
  - Star trackers
- Processors and software needed for:
  - Attitude determination
  - Control

Reaction Wheel





- Enables 3-axis precision pointing
  - Stores torque and moment by the wheel spinning causing counter-rotation of space vehicle because of conservation of momentum
  - Produces torque and momentum
- How it works? To counter a rotation along one of the spacecraft's axes, the reaction wheel…
  - Spins counter to rotation along the wheel's axes is aligned with the axes of the rotation.
  - Reduces angular velocity to zero
- Common configuration is three-axes with the units orthogonally placed.
- Issue: High failure rate:
  - Larger space vehicles can include three backup wheels.
  - Smaller, SWAP limited, vehicles typically use a four-wheel configuration providing one backup-wheel (often misaligned with other wheels)
- If the counter rotation of the reaction wheel becomes saturated:
  - Torque in opposing direction causes wheel rotation to slow, which allows it to store additional momentum and torque.
    - External actuation (e.g. thruster or magnetic torquer)

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Reaction wheel demonstration

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Magnetorquers

Provide control torque motion in the plan perpendicular to the local external magnetic field and magnetic dipole.

- Magnetometer needed to measure local electric field.

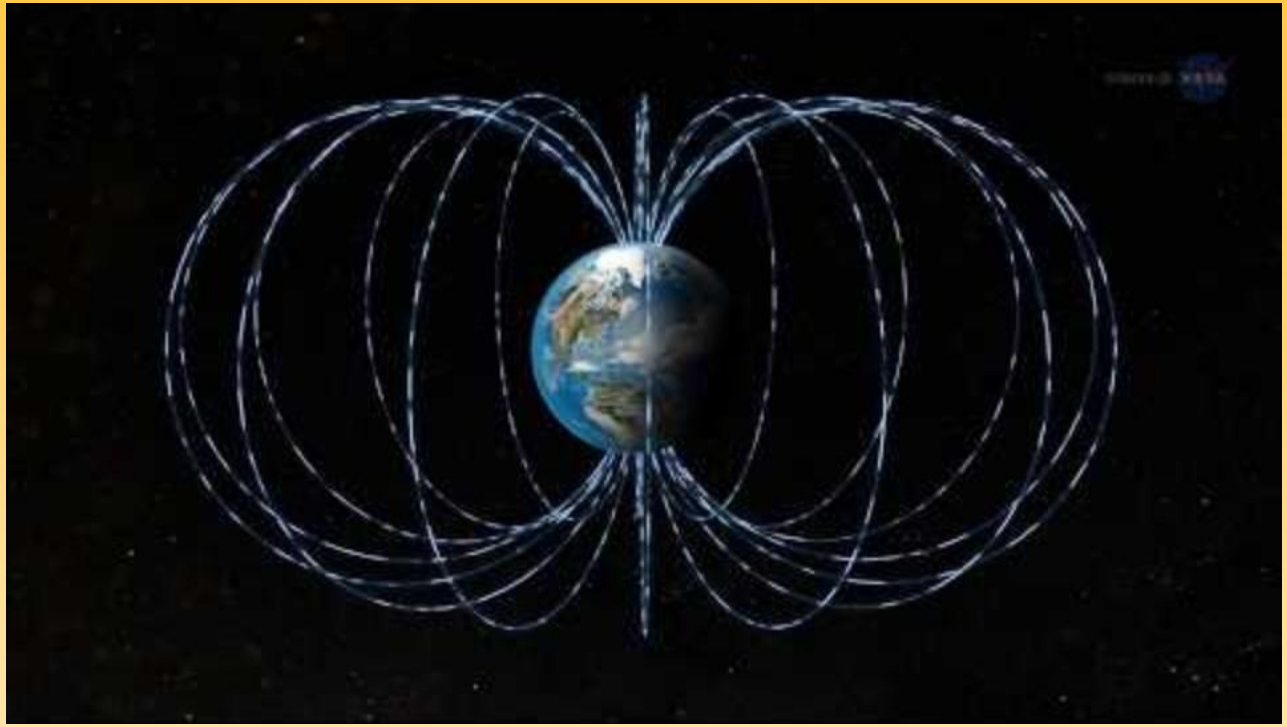
Alternative to using thruster to desaturate spinning wheel for larger satellite systems.

Types:

- Air coils
- Torquer bars/rods

Typically, space vehicle will have three magnetic torquers orthogonal to one another along the x, y, and z axis.

For interplanetary travel, they are less useful due to a lack of a local external magnetic field.



$$\vec{T} = \vec{M} x \vec{B}$$

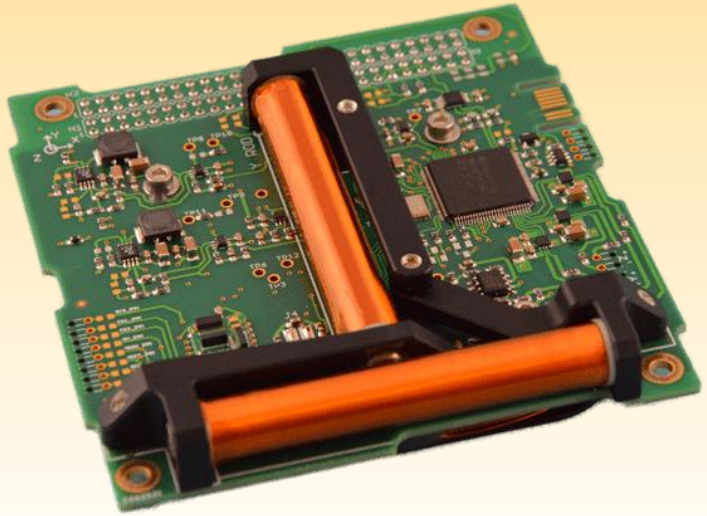
Where,

T is the torque vector (N.m)

M is the magnetic dipole field ($A.m^2$)

m=N.I.A (N turns, I current, A coil area

B is the external magnetic field (Tesla).



ISIS Magnetorquer board (iMTQ) - CubeSatShop.com

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Thrusters

- Enables
  - Attitude
  - Translational control
- Expendable fuel source
- Not typically used for SmallSats and smaller spacecraft
  - Magnetorquer
- Provides large external torque
  - Quick reactions
  - wheel desaturation
  - Spin Stabilization

File:LM RCS thruster module.jpg - Wikimedia Commons

- Proximity operations
  - Higher precision for
    - Translational
    - Attitude controls
  - Attitude control thrusters may provide the best solution.

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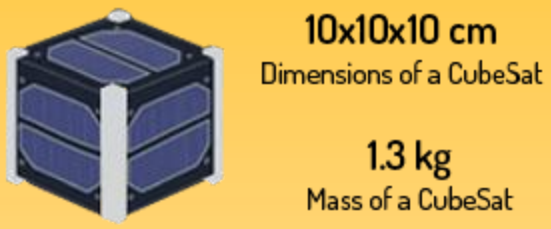
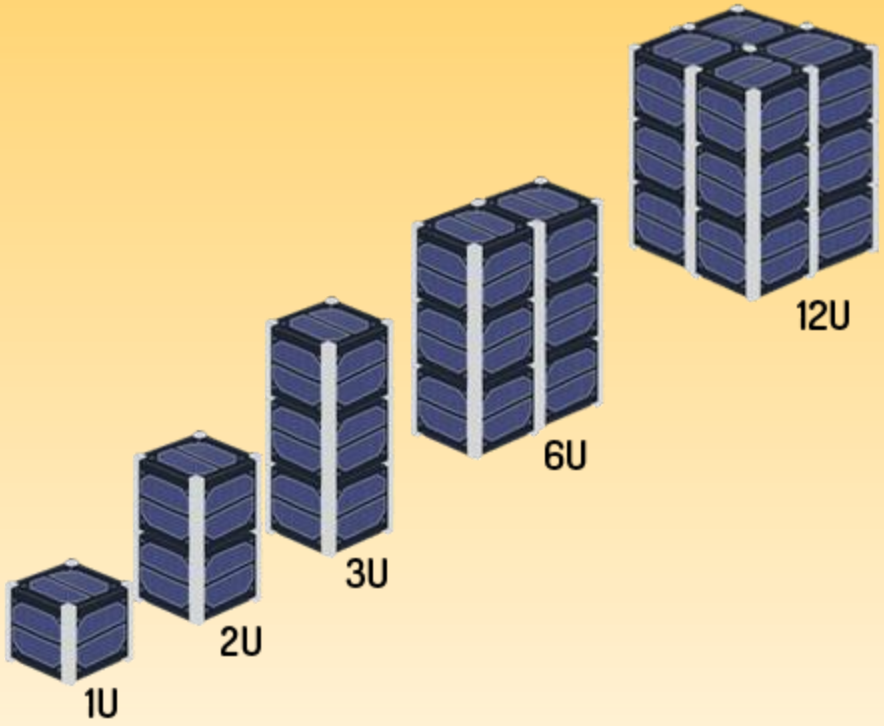
Small Satellite Challenges

- **Miniaturized INS**
  - Angular drift
  - Susceptibility to External Electromagnetic Interference (EMI)
  - Coarse navigation and poor attitude sensing



- **Navigation accuracy**
  - Vulnerability to latency between communication nodes and data links.
  - Fix
    - Data Fusion
    - Fast-response data links (e.g. laser communication links)
  - Low-altitude flight challenges
    - Complex environment
    - GPS-denied environments
    - Resolution
      - Optical stereo vision
      - Star Tracking



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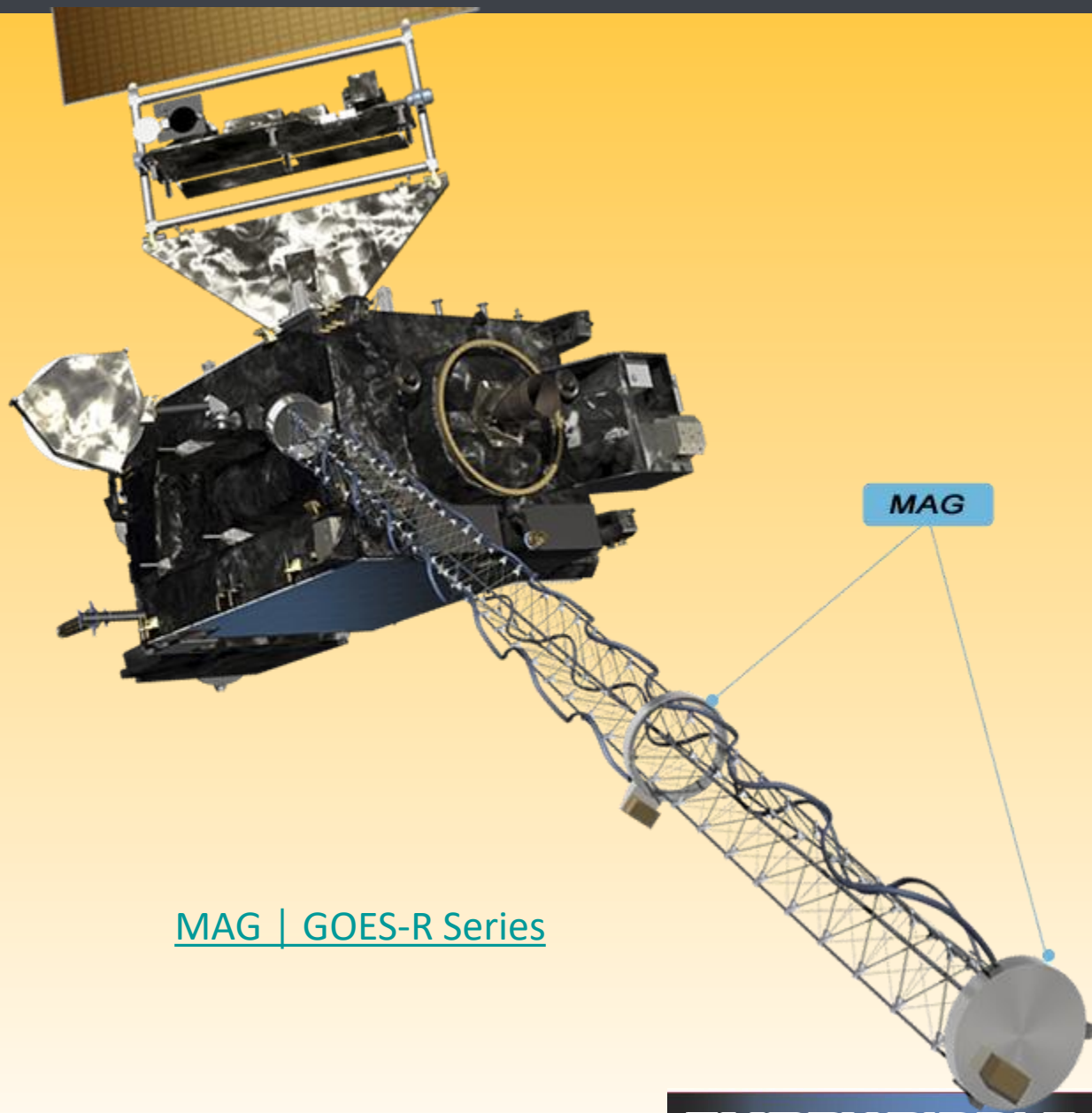
Magnetometers

- Detect and measure local magnetic field.
- Provides real-time 2-axis attitude
- Modeled magnetic field needed to compare with measured field.
- With a time-history of magnetometer data, Kalman filters and similar algorithms can provide
  - Magnetometer only, it can provide
    - Attitude
    - Rotational rates
- Local magnetic field knowledge needed to determine the control of magnetic torquers.
- Challenges
  - Need to avoid interference from local magnetic field sources such as magnetic torquers and reaction wheels.
    - Placement on boom.
  - May require on-orbit recalibration



MAG | GOES-R Series

Sun Sensors

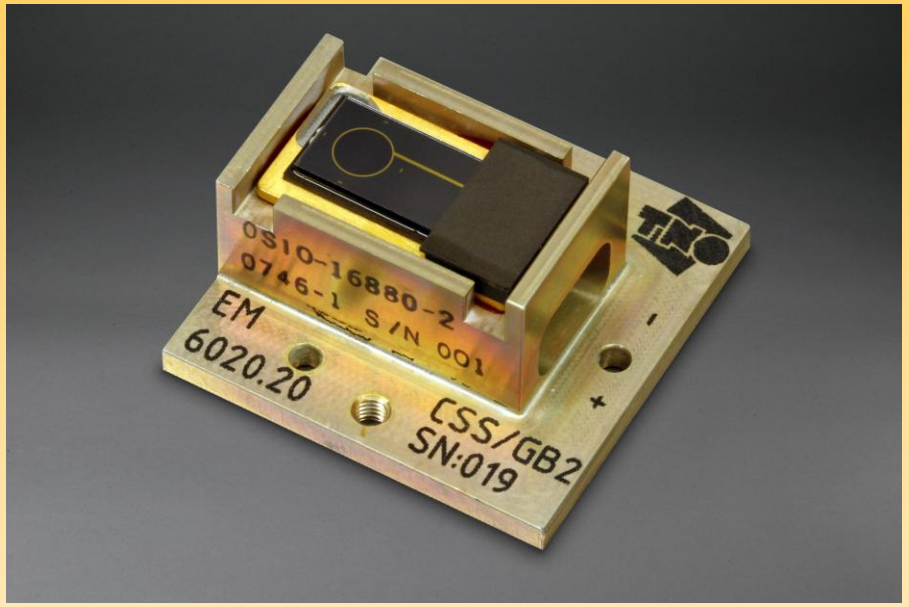
- Enables attitude estimation based on the direction of the Sun
  - Measured with respect to spacecraft's body reference system.
  - 3-axis attitude requires an additional reference
    - Nadir vector

- Useful for fault detection and recovery

- Error in readings
  - Sunlight glinting off other space craft
  - Albedo (diffuse reflection sunlight / solar energy) of the earth or moon.

Types of sun sensors

- Cosine Detectors (aka photocells) aka Coarse Sun Sensors
  - Current generated by cell is proportional to the cosine of the angle between the sensor boresight and the sun.
  - Multiple may be needed to cover full sky.
    - Different locations around spacecraft
  - Advantages
    - Inexpensive
    - Low mass
  - Disadvantages
    - Sensitive to albedo effects
    - Few degree accuracy
    - Requires ADC (analog to digital converter)

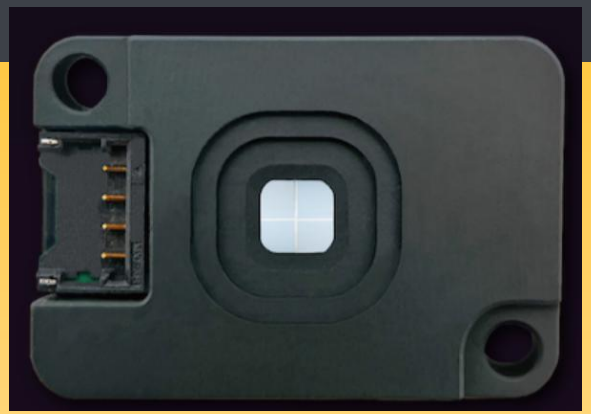


Cosine Sun Sensor | Bradford Space

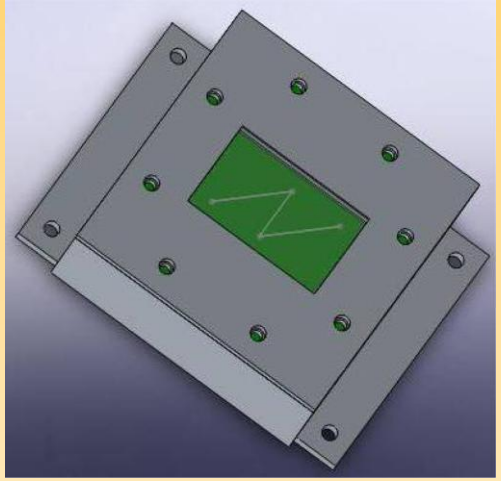
- Albedo – measurement of reflectivity
  - Earth Albedo averages 30%, but varies dramatically based on surface conditions (e.g. snow/ice vs. forests)
  - LEO spacecraft have reduced accuracy up to 10 degrees (coarse sun sensor)
  - Glint of sun off of horizon or other spacecraft can impact some sun sensors' performance

More Sun Sensor Types

- Quadrant detectors
  - Sunlight enters window with 2x2 array of photo diodes
  - Current output is a function of the sun's location within the boresight
  - Also sensitive to Albedo effect



- Digital Sun Sensors
  - Narrow slit with photo diodes behind a "geometric coded bit mask."
  - Mask enables the light to form different measurements based upon the sun's angle
  - Amplified and thresholded signals are then used to derive the sun's angle with respect to the sensor



- Sun Cameras
  - Determine sun position through camera image
  - Optics must be carefully considered due to the bright sunlight.
  - Finds centroid of sun in image.
  - Multiple apertures can increase accuracy
  - Advantages
    - High accuracy
    - Albedo rejection

Star Trackers

- Determine the space vehicle's attitude.
  - Compare positions of celestial bodies relative to an image
  - 50-60 navigational stars identified by astronomers

- Components:
  - Optical Assembly
  - Imaging Sensor
  - Processing Unit and software

- Process for navigation
  - Image capture
  - Star Identification
  - Attitude Calculation
  - Continuous update

- Hybrid navigation systems combine star trackers, gyroscopes, and accelerometers.

- Benefits:
  - High precision (arcsecond-level) attitude determination
  - Enables precise spacecraft pointing
  - Onboard, real-time navigation
  - Applicable across many applications including:
    - Deep space exploration
    - Communication satellite pointing
    - Large and small satellite systems
    - Remote sensing

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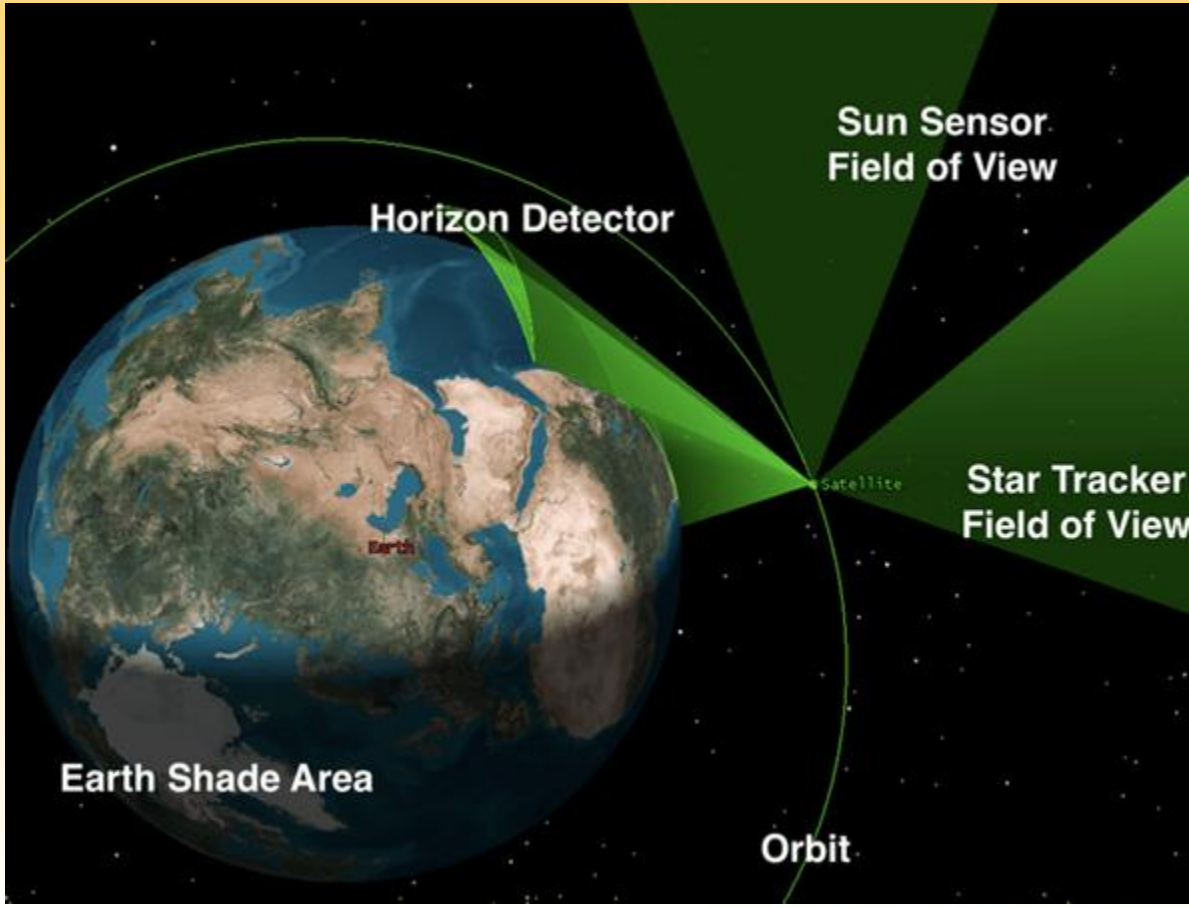
Horizon Sensors

- Types
  - Infrared Horizon Crossing Indicators (HCI)
  - Thermopile sensor
    - Compares temperature difference between the poles and equator.
- Determines
  - Horizon over several measurements
  - Nadir vector
    - Enables attitude determination and guidance



Inertial Sensors

- Elements
  - Gyroscopes
    - Angular velocity
  - Accelerometers
    - Measure velocity change
- Packaging
  - Single Instrument
    - Singly packaged gyro
    - Singly packaged accelerometer
  - Inertial Reference Unity
    - Package with 3 orthogonal gyros and 3 accelerometers.

$$\theta_{axis}(t) = \theta_0 + \int_0^2 \omega_{axis}(t)dt$$

Where,
$\theta(t)$ is current angle
θ_0 is initial angle
$\omega(t)$ is the angular velocity from gyroscope

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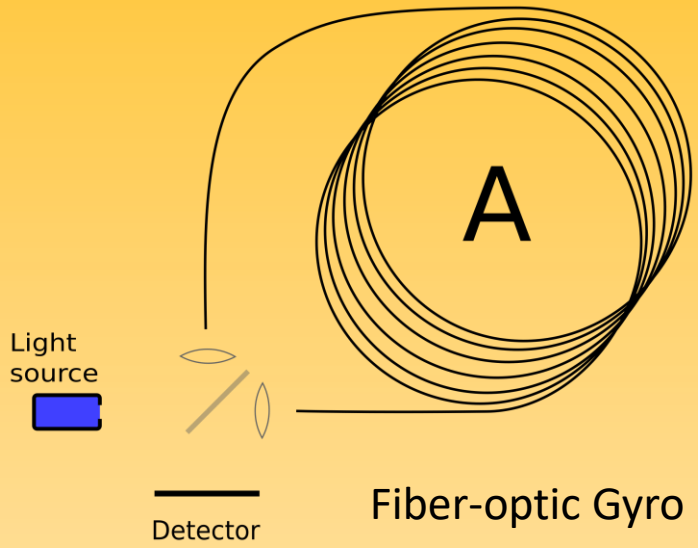
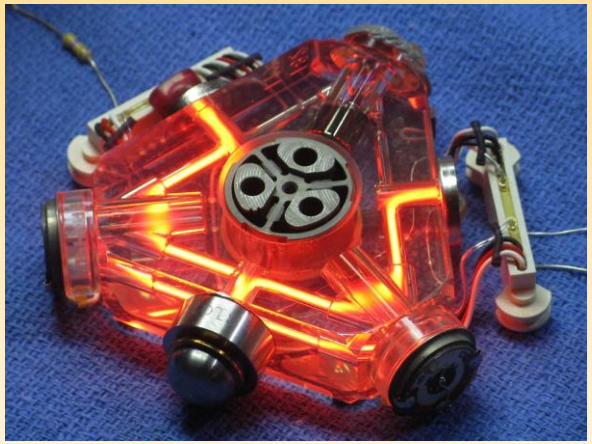
Inertial Sensors (2)

- Types of Gyros
  - Fiber Optic Gyros (FOGs) – exploits the Sagnac Effect
    - Superior performance with higher SWAP requirements
      - SWAP - Size Weight and Power
  - Ring laser gyroscopes
  - MEMS gyros
    - More susceptible to radiation
    - Single event upsets
    - Radiation hardened components are new



Fiber-optic Gyro

- Gyro/Accelerometer stability characterized based on
  - Bias stability
  - Angle random walk
  - Dynamic range
  - Output resolution


Ring laser gyroscope

- Sensor fusion needed to mitigate gyroscope drift
  - Accelerometer – corrects roll and pitch drift given “down” reference
  - Magnetometer – corrects yaw given “North” reference

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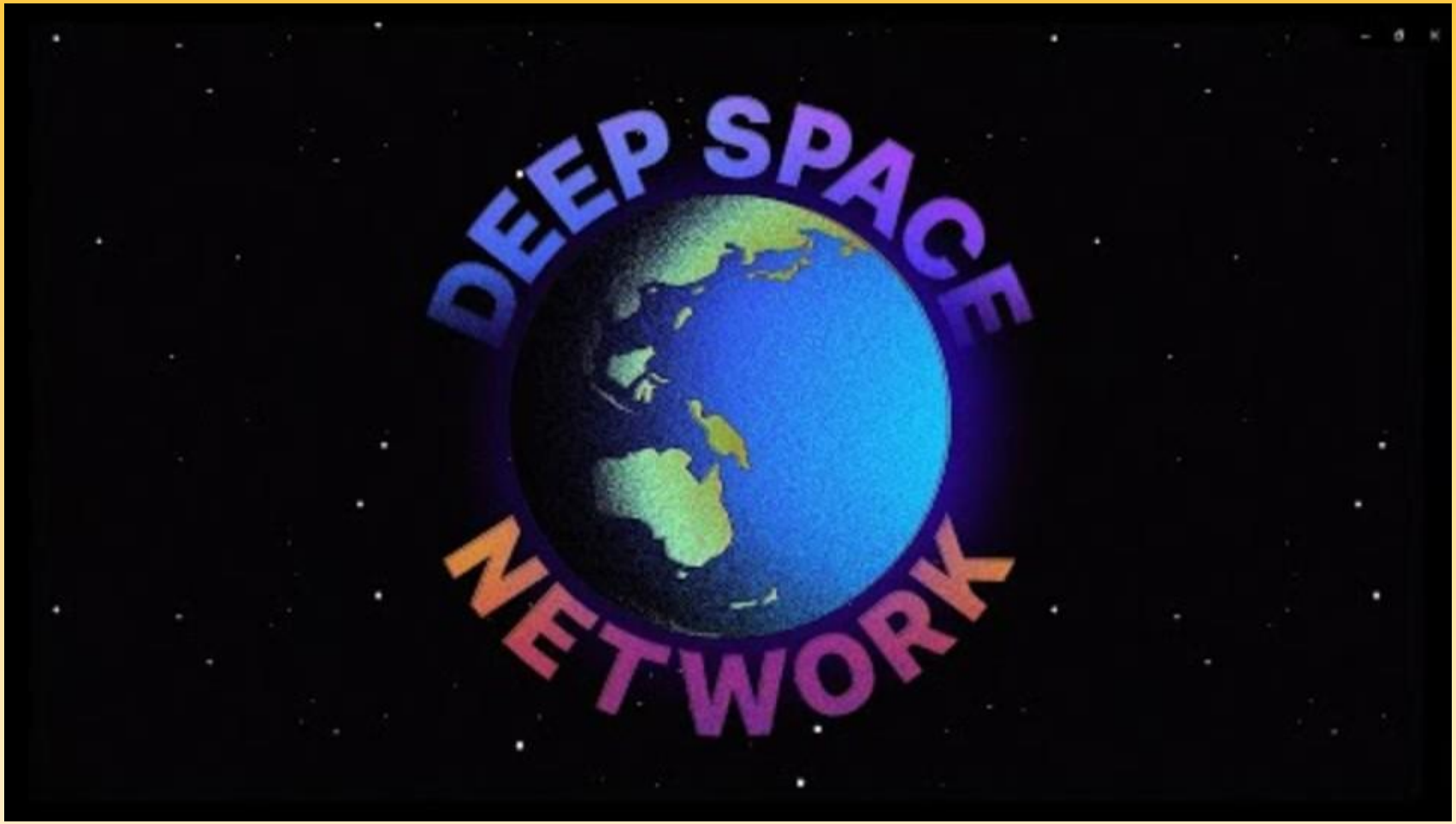
GPS

- See previous slides
- Useful for Low Earth Orbit.
- However, progress made toward GEO and Cislunar potentially on the horizon.

Deep Space Navigation



- Examples:
  - Small Deep Space Transponder (SDST)
    - Radio transponders used in conjunction with the Deep Space Network
    - Designed by JPL, manufactured by General Dynamics
  - IRIS V2 – deep space transponder for smaller form factor spacecraft such as cube sats.

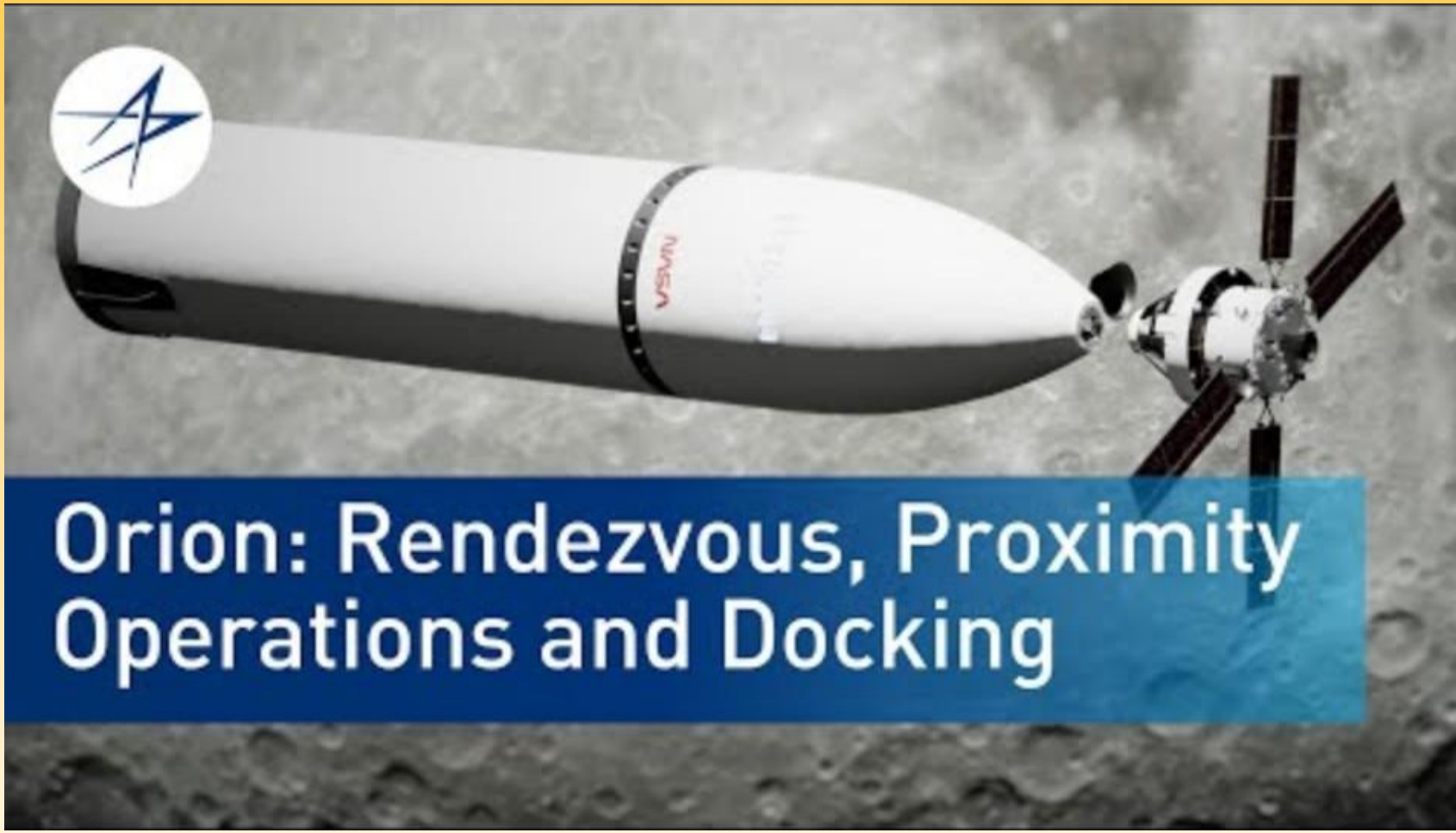
LIDAR

- Light Detection and Ranging (LIDAR)
  - Time-of-flight sensor (laser-object-reflected laser light)
  - Used across many different industries.

- Used for larger satellites doing proximity operations

- Also viable for smaller satellites and cubesats.

- Outside of GNC, LIDAR plays a large role in remote sensing (surface, atmosphere, oceans)



Questions?

- Reference:
  - 5.0 Guidance, Navigation, and Control – NASA from the NASA SmallSat Institute

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